Local Farm & Market Food Availability System

COSC444 Database Systems - Khalifa University - Python | MySQL | Streamlit | LangChain + Gemini

Team: _____ (educational prototype, fictional data)

Problem & objective

Make local food supply visible: farmers publish harvests, markets find and request produce, administrators keep reference data.

Stock is calculated from a dated movement ledger; forecasts never count as stock; allocations can never oversell a lot.

Three roles, one database

Farmer: farms, crop cycles, harvest lots, photos, losses, allocation to requests.

Market buyer: search by crop / emirate / date / quantity, requests, partial delivery confirmation with discrepancy.

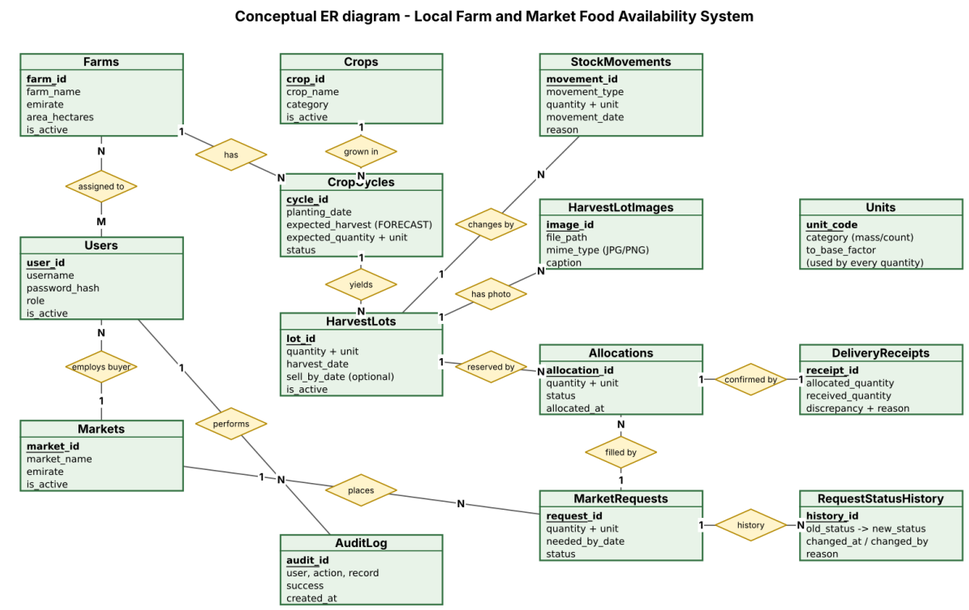
Admin: users, crops, units, markets, approved corrections, analytics dashboard, audit log.

Advanced features

Secure login & audit: bcrypt, lock-out, idle sign-out, ownership rules, append-only audit.

Analytics: supply vs demand, fulfilment & discrepancies, loss & ageing; filters, charts, CSV/PDF export.

Conceptual ER model (14 entities)



LangChain assistant (Gemini)

Ask in plain English; answers come from fixed read-only, role-scoped queries with supporting records shown.

Ambiguous questions get a clarification, unanswerable ones are declined, unsafe ones are refused before reaching the model. No raw SQL, no writes.

Design highlights

MySQL 8: CHECK, UNIQUE, FK, triggers, views.

One transaction per operation; row locks stop overselling.

Explicit units with documented conversion.

Photos on disk + path in SQL; missing files handled.

Testing

91 automated tests pass on real MySQL (rules, concurrency, security, images, GUI).

12 assistant question groups incl. ambiguous & unsafe.

Run it

pip install -r requirements.txt

python init_db.py

streamlit run app.py

Demo password: Demo@1234